

Raghav Senthil Kumar

623-332-2168 | raghavs9@illinois.edu | linkedin.com/in/raghav | github.com/raghav

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2030

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Software Engineer | Disruption Lab @ UIUC

September 2026 - Present

- Built an *MCP* server and developer forum for **AgentPhone (YC S26)** in an *Agile* team of **5** with weekly standups
- Integrated **88 MCP tools** from the API into a *Composio* server and presented the implementation to the client
- Designed **300+ unit test cases** to verify AI agent tool calls and solved a security bug in input path encoding
- Developed an AI workflow with *Jev* to classify support tickets for the dev forum, saving **84% in API costs**

Founding Software Engineer | Clusion

September 2025 – January 2026

- Built a backend search engine (*Python FastAPI*) for a debate SaaS app on *AWS Bedrock* with **10K+ site visitors**
- Designed a *Python* text-retrieval algorithm that extracts evidence segments from **200 GB** of data using URL validation & IQR filtering. Indexed **600K+ documents** on *Elasticsearch*, returning query results in **1 second**
- Developed a de-duplication algorithm that flagged **30%** of evidence documents as identical, reducing index bloat
- Implemented an AI workflow that ran text retrieval weekly & updated the search engine with **1K+ docs**

PROJECTS

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Designed a two-stage *EfficientNetV2-S* computer vision neural network (**4.39M** parameters) in *PyTorch*, trained on **15K+ images** to classify waste into 4 disposal pathways at **96.13% accuracy**, pre-printed research paper on *arXiv*
- Built a synthetic data generation pipeline using *U2-Net* segmentation and alpha-compositing to generate 9K images to train the model to detect contamination in recyclable waste. **Placed 2nd at the Arizona State Science Fair**
- Engineered an embedded robotics prototype: a smart bin on *Arduino Uno* that sorts waste with **2 second second**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Engineered a **suite of 3 edge AI tools** that collects data regarding a school's energy, water and food consumption
- Built a *Python FastAPI* backend that ingests live water, waste & energy data from all 3 tools, runs **4 Random Forest** ML models to detect anomalies & deliver actionable steps via **Gemma Voice LLM** on a *Next.js* dashboard
- Led a team of 5 across project ideation, development and presentation. **Finalist at USAII Global Hackathon**

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Led team of 4 to build a VLM-based pricing algorithm from user-inputted truck images for **Kamion (YC S22)**
- Scraped **15,000** truck-listing images for data mining, auditing quality via de-duplication and non-USD filtering
- Designed a *JSON schema* for a **Claude Sonnet 4.6 agent** to extract condition, brand, model and year from photos
- Revised the schema after each wrong prediction until **10 consecutive images** were correct on each of **3 features**
- Built a *quantile regression* pricing model reaching **87% accuracy**. **2nd Place at UIUC Founder's Hackathon**

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Built a **scheduling agent** (*React/Firebase*) that auto-books meetings for large groups (30+ ppl) on Google Calendar
- Designed a **5-step Python scheduling algorithm** that scans calendar availability, returns a ranked list of time slots and sends email confirmation for each booking. Validated algorithm across **14 unit and integration test cases**
- Created a lobby system that enables the host to authenticate invited participants and queue follow-up meetings
- Licensed Chronos to **8 startups** & signed **2 LOIs worth \$3.5K**. **Won 2025 Congressional App Challenge**

SKILLS

Programming Languages: C++, Python, Java, JavaScript, SQL, HTML/CSS

AI/ML: PyTorch, OpenCV, TensorFlow, Keras, scikit-learn, Hugging Face, Jev, AI Agents, AI Applications

Frameworks/Platforms: FastAPI, Flask, React, Next.js, PostgreSQL, Elasticsearch, Supabase, Firebase, AWS

DevTools: Git, GitHub, Docker, CI/CD, Linux, Clang, Catch2, GDB Debugger, VS Code, Claude Code, Codex

Concepts/Practices: Machine Learning, Computer Vision, Robotics, Software Design, Computer Architecture